
Exercise Set 1 GEOV265 - Tiffany Ajayi

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Question 1: Matlab commands and matrices

% 1a)

```
Matrix = ones(200,300)*9 + eye(200,300)*-13;  
Matrix(1,:)=-4;  
Matrix(200,:)=-4;
```

% 1b)

```
a = [3:6:117].^4;  
b = [-6:-6:-120].^4;  
  
total = sum(a)-sum(b)
```

```
x = [3:3:120].^4;
```

```
s=1;  
t = 0;  
for i =(x)  
    s = s+1;  
    d = (-1)^s;  
  
    t = t+d*i;
```

```
end
```

```
tot = sum(t)
```

```
clear vars
```

```
total =
```

```
-108862380
```

```
tot =
```

-108862380

Question 2: Matlab and Seismology

```
seis1 = rand(20,7);  
seis2 = rand(7,20);
```

% 2a)

```
Ptime = seis1(:,7);  
Pav = mean(Ptime(:));
```

% 2b)

```
Tseis2 = seis2';  
seis3 = [seis1;Tseis2];
```

% 2c)

```
colat = seis3(:,4);  
lat = 90-colat;  
seis3(:,4)= lat;
```

% 2d)

```
last20 = seis3((end-19):end,:);
```

```
radius = ones(20,1)*6371;
```

% taking out the latitude and longitude for the last 20 sources

```
xlat = last20(:,1);  
xlon = last20(:,2);
```

% taking out the latitude and lognitude for the last 20 receivers

```
plat = last20(:,4);  
plon = last20(:,5);
```

% using comando sph2cart to get the cartesians coordinates

```
[xx,yy,zz]=sph2cart(xlon,xlat,radius);
```

```
p1= ([xx,yy,zz])';
```

```
[pp,qq,tt]=sph2cart(plon,plat,radius);
```

```
p2 = ([pp,qq,tt])';
```

```
% computing the dot product and the norm
```

```
epidis = acos(dot(p1,p2)./(norm(p1)*norm(p2)))
```

```
epidis =
```

```
Columns 1 through 7
```

```
1.5266    1.5245    1.5300    1.5156    1.5192    1.5221    1.5166
```

```
Columns 8 through 14
```

```
1.5330    1.5220    1.5321    1.5197    1.5155    1.5256    1.5212
```

```
Columns 15 through 20
```

```
1.5209    1.5240    1.5164    1.5242    1.5168    1.5152
```

2e)

```
first30 = seis3(1:30,:);
```

```
radius1 = ones(30,1)*6371;
```

```
lat1 = first30(:,1);
```

```
lon1 = first30(:,2);
```

```
lat2=first30(:,4);
```

```
lon2 = first30(:,5);
```

```
[x1,y1,z1]=sph2cart(lat1,lon1,radius1);
```

```
p11= ([x1 y1, z1])';
```

```
[x2,y2,z2]=sph2cart(lat2,lon2,radius1);
```

```
p22 = ([x2 y2,z2])';
```

```
epidis1 = acos(dot(p11,p22)./(norm(p11)*norm(p22)));
```

```
r = 6371;
```

```
y = r*sin(epidis1/2);
```

```
k1 = sqrt(r^2-y.^2);
```

```
Delta = 6371-2990;
```

```
k2 = k1-Delta;  
r2 = sqrt(k2.^2+y.^2);  
hyp= 2*r2;  
  
PcP_t = (hyp/10.5)/60  
  
clear vars  
  
PcP_t =  
  
Columns 1 through 7  
  
14.5979    14.5931    14.5855    14.6327    14.5445    14.5550    14.5835  
  
Columns 8 through 14  
  
14.5571    14.5541    14.5891    14.6106    14.5781    14.5382    14.5575  
  
Columns 15 through 21  
  
14.5827    14.5464    14.6190    14.5452    14.5423    14.5445    14.5840  
  
Columns 22 through 28  
  
14.5729    14.5959    14.5423    14.5495    14.5537    14.5485    14.6052  
  
Columns 29 through 30  
  
14.5560    14.6079
```

Question 3: Magma Chamber Dynamics

```
% 3a)  
  
% volume of the source  
vol = 5^3 *1e-9;  
  
% reciever point  
y = [3,2,0.6];  
  
% Source point  
x = [2,-2,-3];  
  
r = norm(x-y);  
  
d = 0.002*vol.*((y(3)-x(3))./r.^3);  
  
displacement = d*1e5  
  
% 3b)
```

```
xrec = 0;
yrec=linspace(-1.875,1.875,31);
zrec = 0;

disp = zeros(size(yrec));

for i = 1:length(yrec)

    r1 = (xrec - x(1))^2 + (yrec(i) - x(2)).^2 + (zrec - x(3))^2;

    dist =sqrt(r1);

    disp(i) = 0.002*vol * (zrec-x(3))./dist.^3;
end

figure(1)

plot(yrec, disp*1e5, 'LineWidth',2)
grid on
ylabel('Vertical Displacement in [cm]')
xlabel('Receiver Distance in [km]')
title('Vertical Displacement due to a point source')
```

```
clear vars
```

```
% 3c)
```

```
x = [-1:0.005:1];
y = [-0.5:0.005:1.7];
z = -3;

vol = 5^3 *1e-9;

origin=[0 0 0];
yrec=linspace(-1.875,1.875,31);

[X,Y]=meshgrid(x,y);

vertical_displacement = zeros(31,1);
disp = zeros(length(y), length(x));

for i = 1:length(yrec)

    dist1 = (X-origin(1)).^2 + (Y-yrec(i)).^2 + (z-origin(3)).^2;

    r = sqrt(dist1).^3;
    disp = (origin(3)-z)./r;

    vertical_displacement(i) = 1e5*vol*0.002*sum(disp(:));
end
```

```
figure(3)
plot(yrec,vertical_displacement)
grid on
ylabel('Vertical Displacement in [cm]')
xlabel('Receiver Distance in [km]')
title('Vertical Displacement due to a plane volume source')

clear var

% 3d)
cir = [0:18:342];

r = 0.3; % diameter gitt som 600m, radius = 300m

x = r * cosd(cir);
y = r * sind(cir);
z = zeros(size(x));

% 3e)

sx = [-1:0.005:1];
sy = [-0.5:0.005:1.7];
sz = -3;

[SX, SY]=meshgrid(sx,sy);

v = 5^3 *1e-9;
con = 0.002;

dv = zeros(1, length(x));

for i =1:length(x)

    dist = sqrt((SX - x(i)).^2 + (SY - y(i)).^2 + sz^2);

    disp = ((0 - sz)./ dist.^3);
    disp = sum(disp(:));

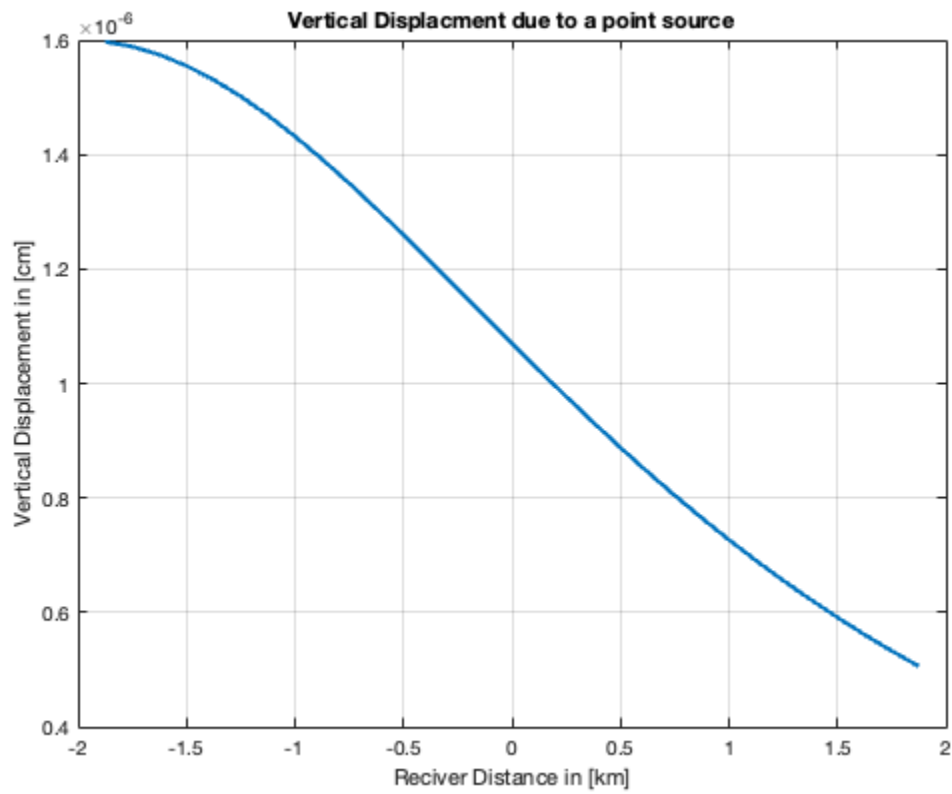
    dv(i) = 1e5*v*con*disp;

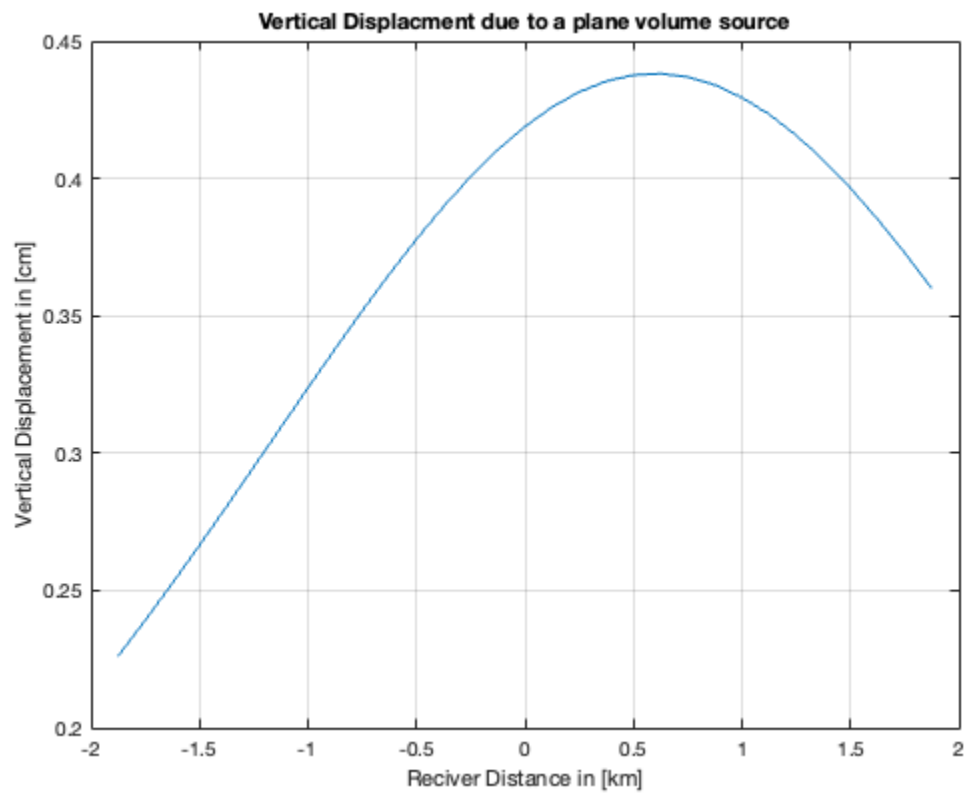
end

% figure(2)
%
%     plot(cir,dv,'*-','LineWidth',2)
%     xlabel('[Degrees]')
%     ylabel('Vertical displacement in cm')
%     title('Vertical Displacement at the circle of receivers')
%
% grid on
```

displacement =

5.4882e-07





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